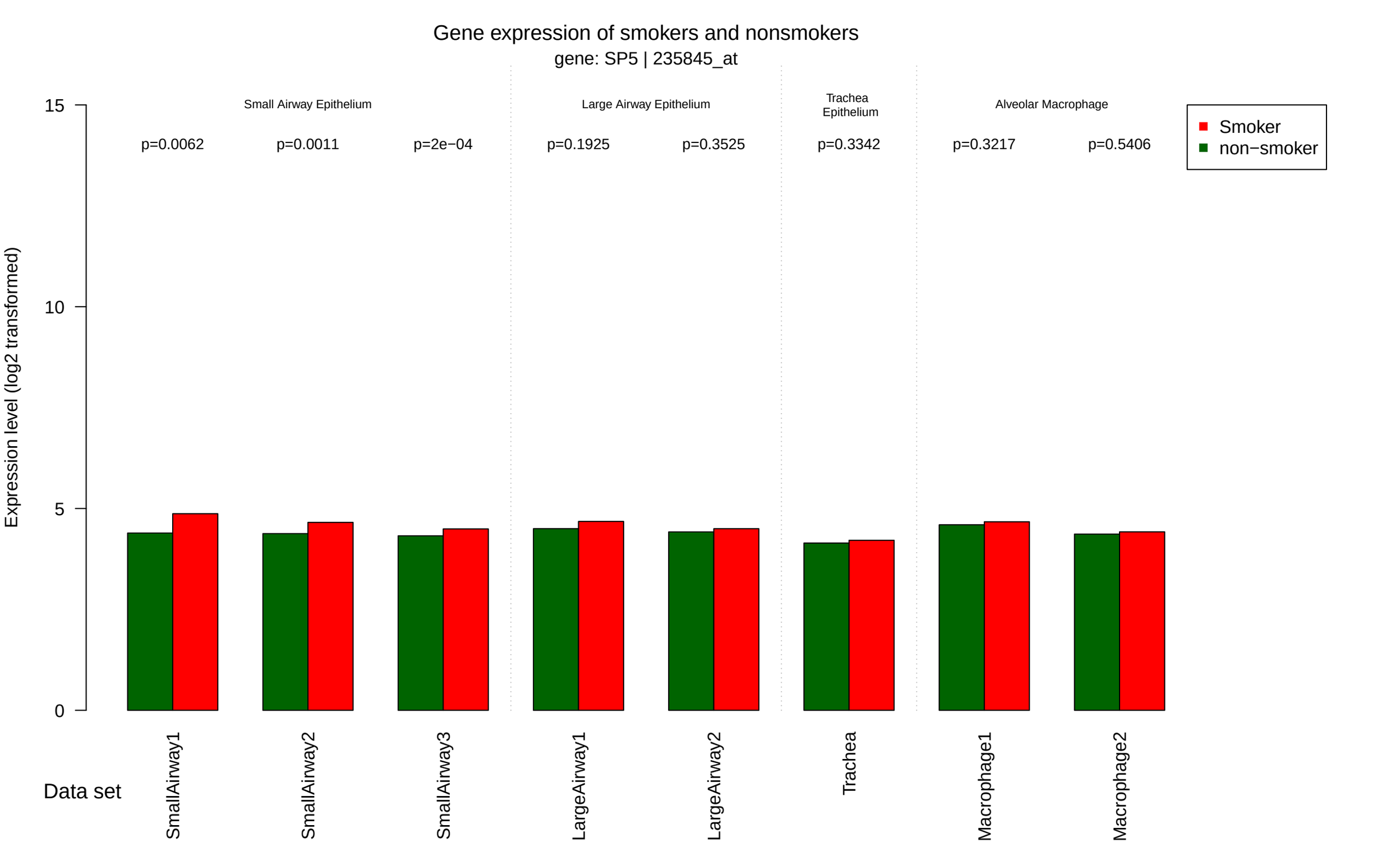
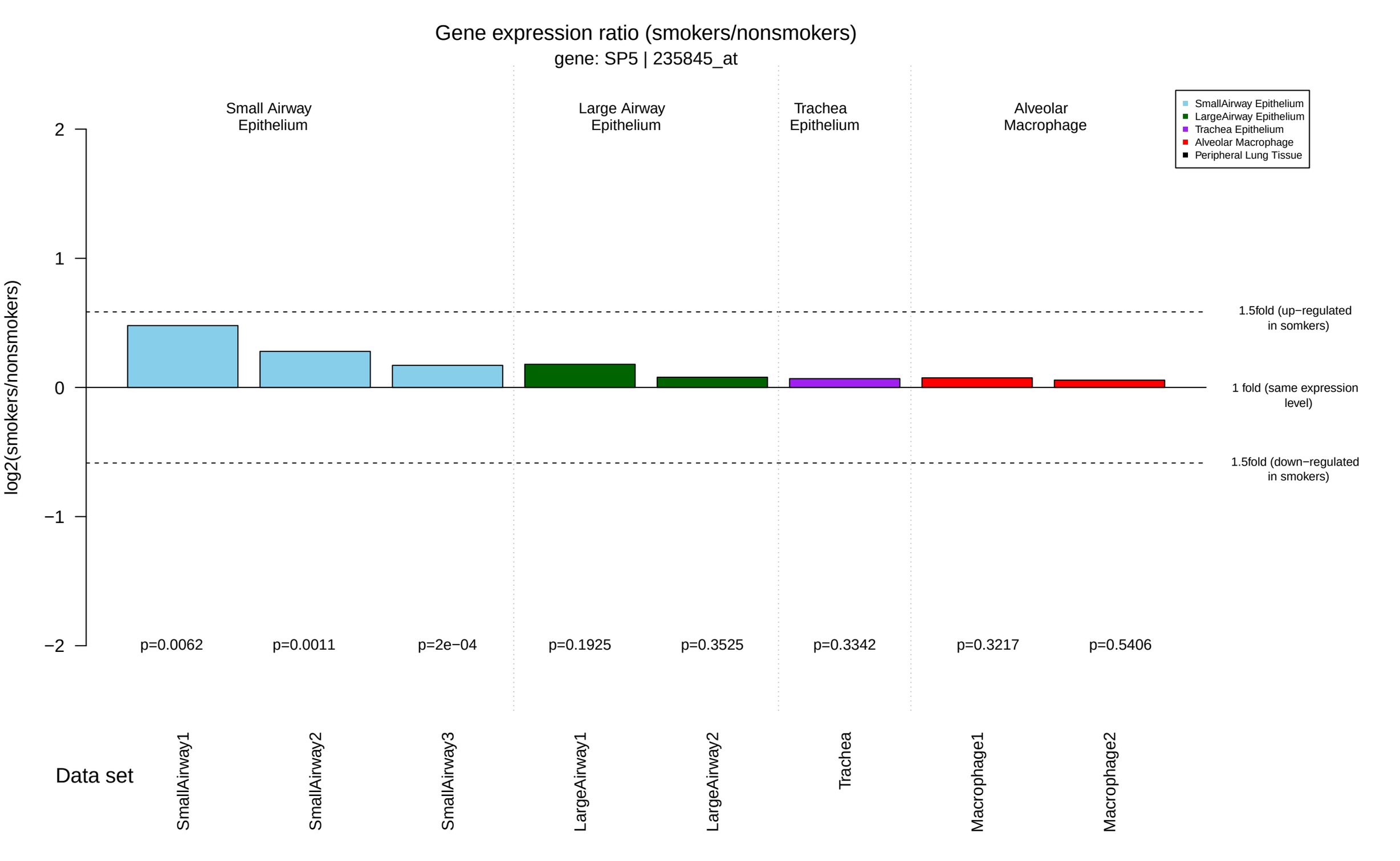
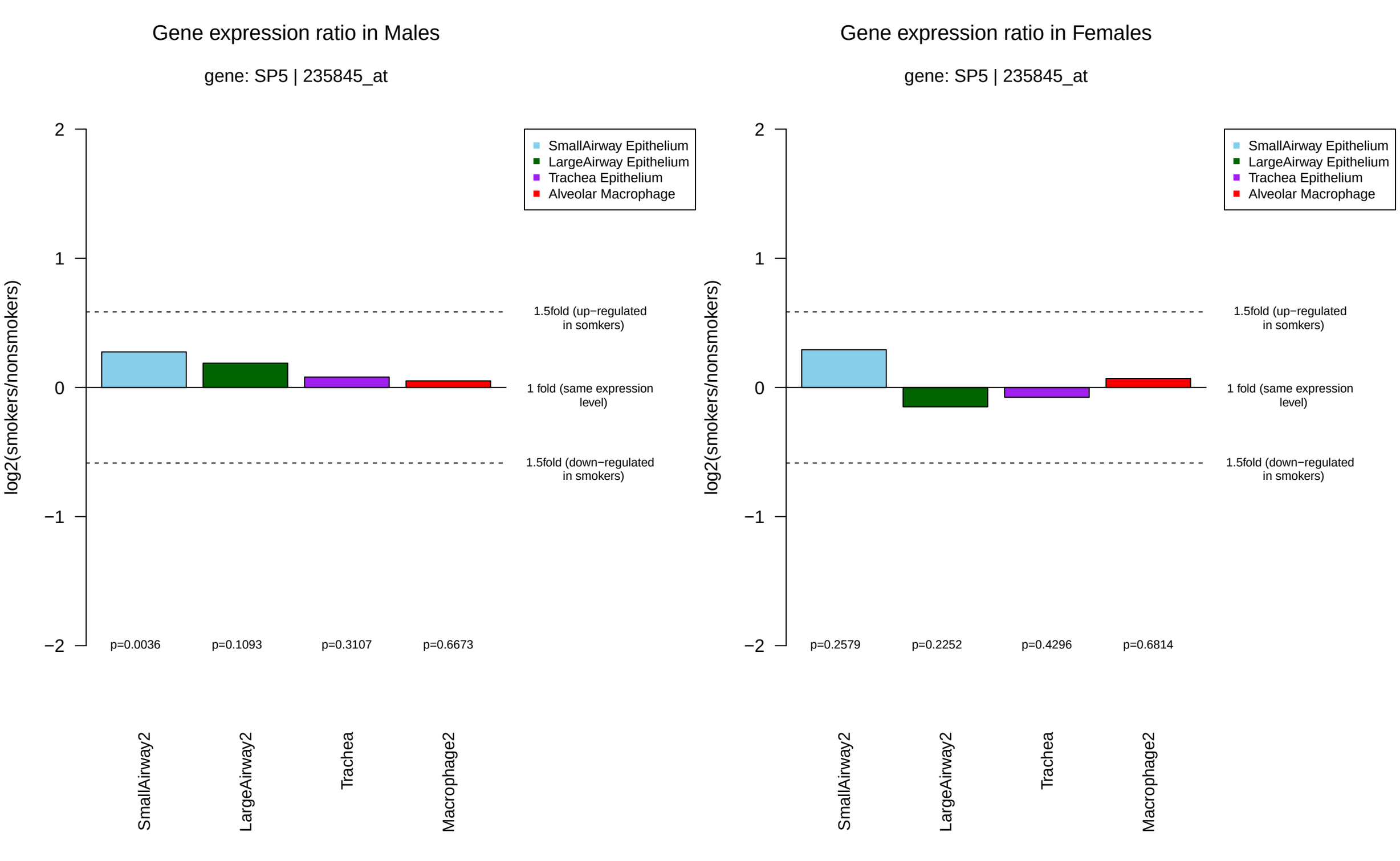
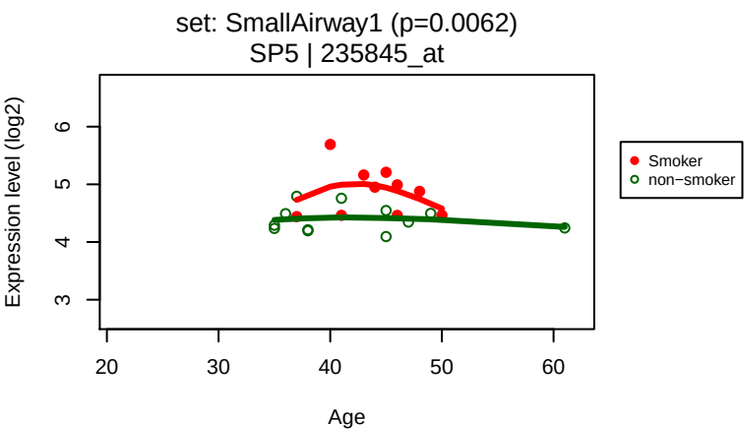
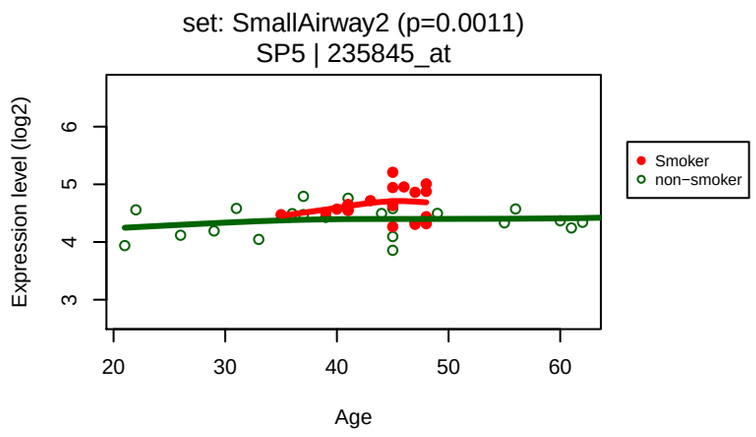
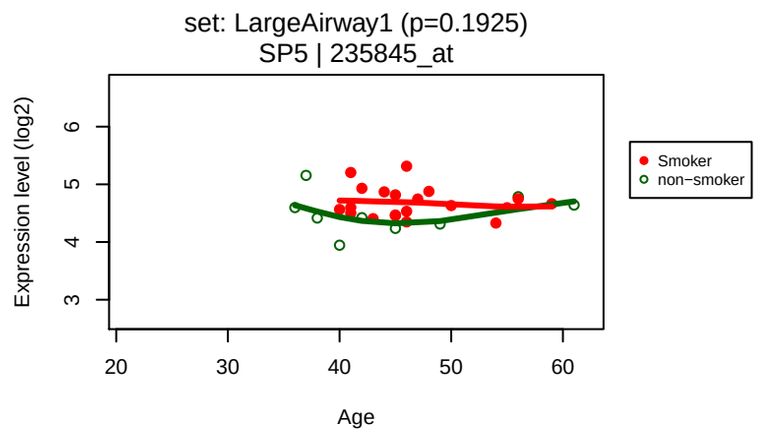

Gene expression of smokers and nonsmokers
gene: SP5 | 235845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0062
p=0.0011
p=2e−04
p=0.1925
p=0.3525
p=0.3342
p=0.3217
p=0.5406
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

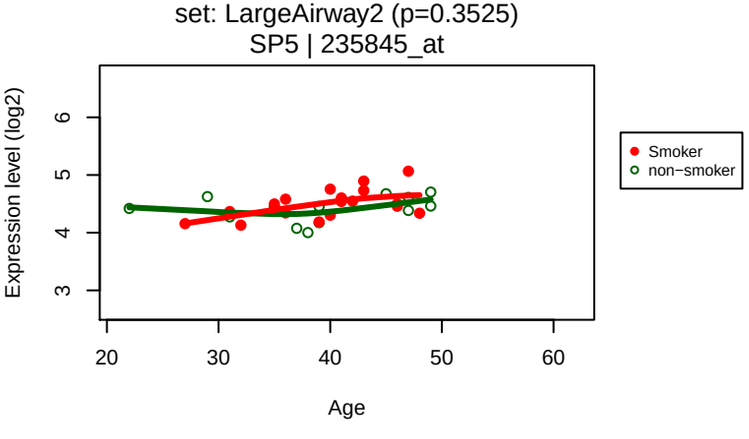

Gene expression ratio (smokers/nonsmokers)
gene: SP5 | 235845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0062
p=0.0011
p=2e−04
p=0.1925
p=0.3525
p=0.3342
p=0.3217
p=0.5406
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

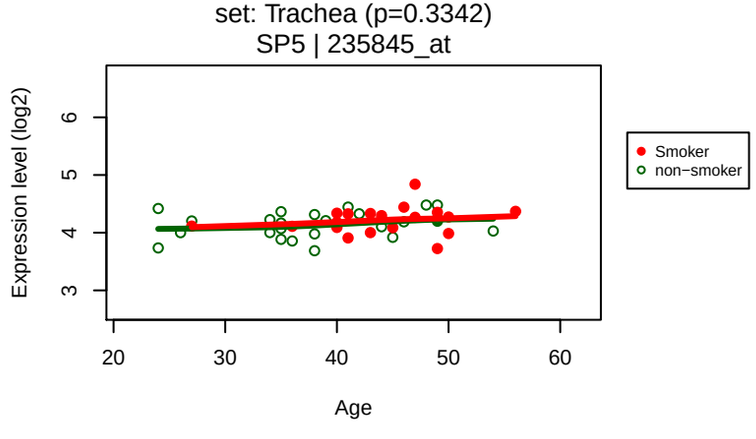

Gene expression ratio in Males
gene: SP5 | 235845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0036
p=0.1093
p=0.3107
p=0.6673
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SP5 | 235845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2579
p=0.2252
p=0.4296
p=0.6814
SmallAirway2
LargeAirway2
Trachea
Macrophage2

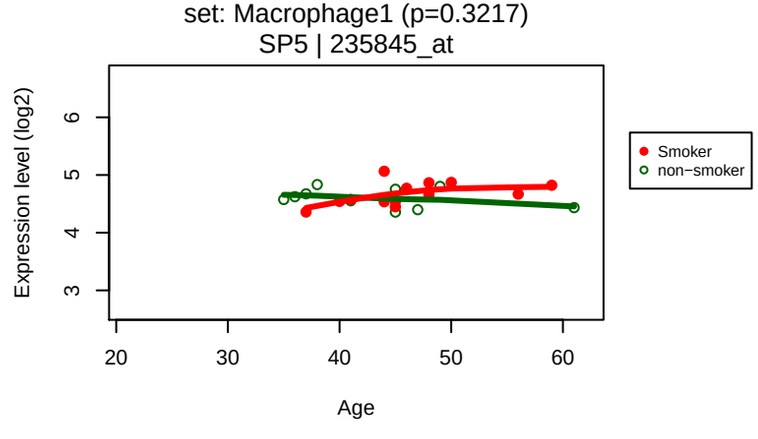

set: SmallAirway1 (p=0.0062)
SP5 | 235845_at
Expression level (log2)
Age
Smoker
non−smoker

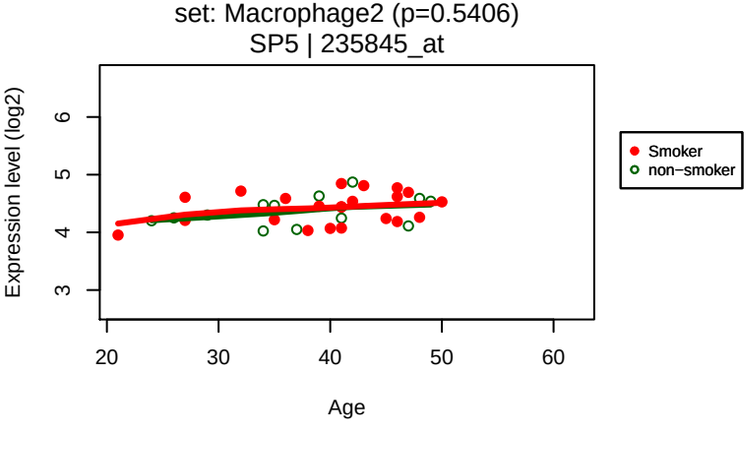

set: SmallAirway2 (p=0.0011)
SP5 | 235845_at
Expression level (log2)
Age
Smoker
non−smoker

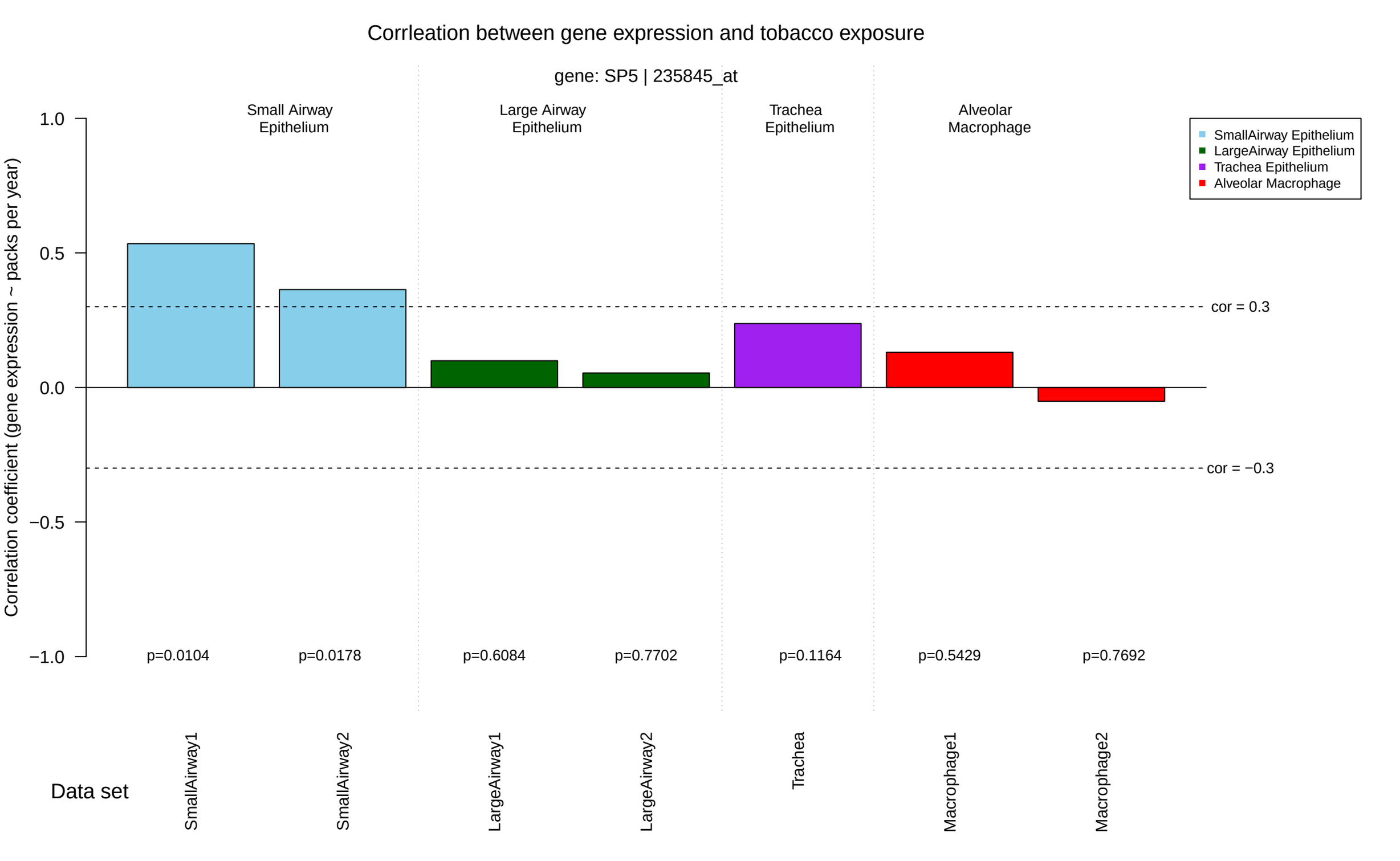

set: LargeAirway1 (p=0.1925)
SP5 | 235845_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3525)
SP5 | 235845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3342)
SP5 | 235845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3217)
SP5 | 235845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5406)
SP5 | 235845_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SP5 | 235845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0104
p=0.0178
p=0.6084
p=0.7702
p=0.1164
p=0.5429
p=0.7692
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2